

Inductive limits

(EXERCISES FOR LECTURES 7–8)

8.1. Let (X_i, φ_{ij}) be an inductive system of locally convex spaces indexed by a directed set I . Let

$$X = \left(\bigoplus_{i \in I} X_i \right) / \text{span}\{x - \varphi_{ij}(x) : i \leq j, x \in X_i\}.$$

For each $j \in I$, let $\varphi_j: X_j \rightarrow X$ denote the composite of the standard embedding $X_j \rightarrow \bigoplus_{i \in I} X_i$ and the quotient map $\bigoplus_{i \in I} X_i \rightarrow X$. Show that (X, φ_i) is a (category-theoretic) inductive limit of (X_i, φ_{ij}) in LCS.

8.2 (*directed families of subspaces*). Let X be a vector space, and let $(X_i)_{i \in I}$ be a family of vector subspaces of X indexed by a directed set I in such a way that $X_i \subset X_j$ whenever $i \leq j$. Let $\varphi_{ij}: X_i \hookrightarrow X_j$ denote the respective inclusion maps.

(a) Show that $Y = \bigcup_i X_i$ together with the inclusion maps $(\varphi_i: X_i \hookrightarrow Y)_{i \in I}$ is an inductive limit of (X_i, φ_{ij}) in the category of vector spaces.

(b) Suppose also that each X_i is equipped with a locally convex topology in such a way that the inclusions φ_{ij} are continuous for all $i \leq j$. Equip Y with the inductive locally convex topology generated by the inclusion maps $\varphi_i: X_i \hookrightarrow Y$. Show that Y together with $(\varphi_i)_{i \in I}$ is an inductive limit of (X_i, φ_{ij}) in LCS.

8.3. Let $F = (X_i, \varphi_{ij})$ be an inductive system of vector spaces indexed by a directed set I , let $X = \varinjlim X_i$, and let $\varphi_i: X_i \rightarrow X$ denote the canonical maps. Show that, for each $i \in I$, we have $\text{Ker } \varphi_i = \bigcup_{j \geq i} \text{Ker } \varphi_{ij}$. Deduce that, if all φ_{ij} are injective, then F is isomorphic (as an inductive system) to a suitable directed family of subspaces of X (see Exercise 8.2 (a)).

8.4. Let K be a compact subset of $\mathbb{C}^n$, and let $\mathcal{U}$ be a base of relatively compact open neighborhoods of K . Recall (see the lecture) that the space $\mathcal{O}(K)$ of germs of holomorphic functions on K is the locally convex inductive limit $\varinjlim \{\mathcal{O}(U) : U \in \mathcal{U}\}$. Construct a topological isomorphism $\mathcal{O}(K) \cong \varinjlim \{\mathcal{A}(\overline{U}) : U \in \mathcal{U}\}$, where $\mathcal{A}(\overline{U})$ is defined in Exercise 6.9.

8.5. Let $\mathbb{D}_r = \{z \in \mathbb{C} : |z| < r\}$. Show that the restriction map $\mathcal{O}(\mathbb{D}_R) \rightarrow \mathcal{O}(\mathbb{D}_r)$ (where $R > r$) is not topologically injective. As a corollary, the inductive sequence $\{\mathcal{O}(\mathbb{D}_{1/n}) : n \in \mathbb{N}\}$ is not strict.

8.6. Prove that the following spaces are Hausdorff: (a) $C_c(X)$ (where X is a topological space); (b) $C_c^\infty(U)$ (where U is an open subset of $\mathbb{R}^n$); (c) $\mathcal{O}(K)$ (where K is a compact subset of $\mathbb{C}^n$). (In (a) and (b), try to avoid using general results on strict inductive limits. In (c), the respective inductive limit is not strict by Exercise 8.5.)

8.7. Given $z \in \mathbb{C}$, let $\mathcal{O}_z$ denote the space of germs of holomorphic functions at z (i.e., $\mathcal{O}_z = \mathcal{O}(\{z\})$, see Exercise 8.4).

(a) Let P denote the set of all sequences $p = (p_n)$, $p_n > 0$, such that $p_n = o(\varepsilon^n)$ for every $\varepsilon > 0$. For each $p \in P$, define a norm $\|\cdot\|_p$ on $\mathcal{O}_z$ by $\|f\|_p = \sum_n |c_n(f)| p_n$, where $c_n(f) = f^{(n)}(z)/n!$. Show that the family $\{\|\cdot\|_p : p \in P\}$ of norms is defining for $\mathcal{O}_z$.

(b) Show that a sequence (f_n) converges in $\mathcal{O}_z$ iff there is a neighborhood $U \ni z$ such that (f_n) is contained in $\mathcal{O}(U)$ and converges uniformly on U .

(c) Is $\mathcal{O}_z$ metrizable?

8.8. Let $\mathcal{C}_0 = \varinjlim C[-1/n, 1/n]$ be the space of germs of continuous functions at $0 \in \mathbb{R}$ equipped with the respective inductive locally convex topology, and let $E = \{f \in \mathcal{C}_0 : f(0) = 0\}$. Prove that the topology on E induced from $\mathcal{C}_0$ is anti-discrete. As a corollary, $\mathcal{C}_0$ is not Hausdorff.

8.9. Let $\mathcal{O}_0$ denote the space of germs of holomorphic functions at $0 \in \mathbb{C}$. Write $\mathcal{O}_0$ in the form $\mathcal{O}_0 = \varinjlim \mathcal{O}(\mathbb{D}_{1/n})$, where $\mathbb{D}_{1/n} = \{z \in \mathbb{C} : |z| < 1/n\}$. Let us now change the canonical topology on $\mathcal{O}_0$ as follows: equip each $\mathcal{O}(\mathbb{D}_{1/n})$ with the topology of pointwise convergence, and equip $\mathcal{O}_0$ with the respective inductive locally convex topology. Let $E = \{f \in \mathcal{O}_0 : f(0) = 0\}$. Prove that the topology on E induced from $\mathcal{O}_0$ is anti-discrete. As a corollary, $\mathcal{O}_0$ is not Hausdorff for the above (nonstandard) topology.

8.10. Let X be the inductive limit of a strict inductive sequence (X_n) of locally convex spaces. Suppose that each X_n is closed in X_{n+1} . Prove that $B \subset X$ is bounded iff there exists n such that $B \subset X_n$ and such that B is bounded in X_n .

8.11. Let Y be a normed space, and let (Y_n) be a strictly increasing sequence of vector subspaces of Y such that Y_n is dense in Y_{n+1} for each n . Let $Z = \bigcup_n Y_n$.

(a) Give a concrete example of such a sequence (Y_n) .

(b) Describe (as explicitly as you can) the inductive locally convex topology on Z generated by the inclusion maps $Y_n \hookrightarrow Z$.

(c) Does the result of Exercise 8.10 hold without the assumption that each X_n is closed in X_{n+1} ?

(d) Suppose that X and X_n are as in Exercise 8.10, and that $X_n \neq X_{n+1}$ for each n . Recall (see the lecture) that X is nonmetrizable. Does this result hold without the assumption that each X_n is closed in X_{n+1} ?